

MORE THAN ENOUGH MATH TO DO BIG O

If your algebra is rusty, here's more than enough math to do big O analysis:

Multiplication Repeated addition. $2 \times 4 = 8$, just like $2 + 2 + 2 + 2 = 8$. With variables, $n + n + n$ is $3 \times n$.

Multiplication notation Algebra notation often omits the $\times$ sign, so $2 \times n$ is written as $2n$. With numbers, 2×3 is written as $2(3)$ or simply 6.

The multiplicative identity property Multiplying a number by 1 results in that number: $5 \times 1 = 5$ and $42 \times 1 = 42$. More generally, $n \times 1 = n$.

The distributive property of multiplication $2 \times (3 + 4) = (2 \times 3) + (2 \times 4)$. Both sides of the equation equal 14. More generally, $a(b + c) = ab + ac$.

Exponentiation Repeated multiplication. $2^4 = 16$ (pronounced “2 raised to the 4th power is 16”), just like $2 \times 2 \times 2 \times 2 = 16$. Here, 2 is the *base* and 4 is the *exponent*. With variables, $n \times n \times n \times n$ is n^4 . In Python, we use the `**` operator: `2 ** 4` evaluates to 16.

The 1st power evaluates to the base $2^1 = 2$ and $9999^1 = 9999$. More generally, $n^1 = n$.

The 0th power always evaluates to 1 $2^0 = 1$ and $9999^0 = 1$. More generally, $n^0 = 1$.

Coefficients Multiplicative factors. In $3n^2 + 4n + 5$, the coefficients are 3, 4, and 5. You can see that 5 is a coefficient because 5 can be rewritten as $5(1)$ and then rewritten as $5n^0$.

Logarithms The inverse of exponentiation. Because $2^4 = 16$, we know that $\log_2(16) = 4$. We pronounce this “the log base 2 of 16 is 4.” In Python, we use the `math.log()` function: `math.log(16, 2)` evaluates to 4.0.

Calculating big O often involves simplifying equations by combining like terms. A *term* is some combination of numbers and variables multiplied together: in $3n^2 + 4n + 5$, the terms are $3n^2$, $4n$, and 5. *Like terms* have the same variables raised to the same exponent. In the expression $3n^2 + 4n + 6n + 5$, the terms $4n$ and $6n$ are like terms. We could simplify and rewrite this as $3n^2 + 10n + 5$.

Keep in mind that because $n \times 1 = n$, an expression like $3n^2 + 5n + 4$ can be thought of as $3n^2 + 5n + 4(1)$. The terms in this expression match with the big O orders $O(n^2)$, $O(n)$, and $O(1)$. This will come up later when we're dropping coefficients for our big O calculations.

These math rule reminders might come in handy when you're first learning how to figure out the big O of a piece of code. But by the time you finish “Analyzing Big O at a